Low-Cost Substrate Mixer and Portioning Device

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Background

Edible mushrooms are widely recognized as a nutritional "superfood" due to their high protein content, essential B-vitamins, and minerals such as potassium, phosphorus, and iron (Oei, 2003, pp.4-5). This nutrient density makes mushrooms a critical supplement to staple-based diets in resource-limited settings, supporting efforts to combat malnutrition (Akter et al., 2022). Beyond their nutritional value, mushrooms represent a cornerstone of the circular economy; they are cultivated on agricultural byproducts such as straw, cotton husks, or coffee waste, that are frequently considered organic waste (Grimm & Wösten, 2018; Oei & van Nieuwenhuijzen, 2005). This bioconversion process reduces environmental pollution and transforms low-value residues into high-quality food and potent organic fertilizer (Grimm & Wösten, 2018). Consequently, mushroom farming is an ideal agricultural option for low-income countries because it provides high nutritional and financial value per unit area with minimal land requirements (Khan et al., 2024).

The technical process of cultivation begins with substrate preparation, where agricultural residues must be chopped, moistened to an optimal target content, and thoroughly mixed. The substrate is then filled into bags or containers and pasteurized or sterilized to suppress competitive microbes (Oei, 2003, pp.19-26). Once cooled, the substrate is inoculated with mushroom mycelium pre-grown on grains, the so-called "spawn". Following an incubation period of approximately two to six weeks, the mycelium fully colonizes the substrate, and fruiting bodies emerge for harvest (Oei & van Nieuwenhuijzen, 2005, p.34).

In the Ugandan context, the potential for this industry is significant, as market demand frequently exceeds supply by over three times. However, production is severely constrained by the intensive manual labour required. As highlighted in correspondences with Samson Kisseka of "*Hello Mushrooms U Ltd*" in Kampala, the intensive labour-input requirement is the primary barrier limiting the growth capacity of smallholder farmers. Currently, substrate mixing, wetting, and bag filling are predominantly performed by hand, which is both time-consuming and physically demanding, and leads to high labour costs (S. Kisseka, personal communication, 2025).

This manual approach presents several technical challenges: agricultural residues like straw often have heterogeneous particle sizes, making uniform mixing difficult (Oei, 2003, pp.89-90). Furthermore, water uptake by dry substrate can be slow and uneven, leading to inconsistent moisture levels that jeopardize the sterilization and growth process. Finally, achieving uniform portioning and bag density by hand is challenging, yet critical for consistent yields and efficient pasteurization (Suwannaeach et al., 2022).

While industrial mechanization solutions exist, they are often prohibitively expensive or unsuitable for the small-scale infrastructure of low-income settings (Higgins et al., 2017). There is a critical need for low-cost, robust, and open-source equipment that can be maintained locally (S. Kisseka, personal communication, 2025). Therefore, this project aims to develop a hand-driven substrate mixer and portioning device specifically designed to reduce the labour cost for smallholder mushroom producers in settings like Uganda, thereby enabling a breakthrough in production efficiency and livelihood sustainability.

Problem statement and goals

Despite the high demand for mushrooms in Uganda, small-scale producers are constrained by the physical and technical limits of manual substrate preparation. Manual mixing often results in uneven water distribution and heterogeneous substrate density, which directly increases the risk of contamination and leads to inconsistent yields. Furthermore, the high labour intensity of hand-filling bags creates a production bottleneck that prevents farmers from scaling their operations to meet market needs. Existing industrial machinery is economically inaccessible and difficult to maintain in low-resource settings, leaving a gap for affordable, locally adaptable solutions. To address this, the goal of this project is to design and prototype a low-cost, hand-driven device that integrates substrate mixing, wetting, and portioning. By providing open-source construction plans, this project aims to empower smallholders like Samson Kisseka to increase efficiency and secure a sustainable income through standardized production.

Research Questions

1. “**Mixing**”

How can substrates like cotton husks, sawdust, shredded corncobs, chopped cereal straws and other agricultural waste be uniformly mixed, so that after a certain stirring time, samples of equal size from three different regions of the agitator are no longer visually distinguishable?

2. “**Watering**”

How can the mixed dry substrate be moistened to achieve a water content of 60-70 wt%?

3. “**Portioning**”

How can the mixed and moistened substrate be filled into bags so that deviations in weight and volume of portions are within a tolerance range of ±10%?

Methods

Handling Uncertainties

This project is subject to little uncertainty, as numerous industrial and self-built substrate mixers already exist and can serve as reliable reference systems. Components with well-defined requirements and low technical risk are therefore developed using a linear, waterfall-like approach. In contrast, subsystems characterized by higher uncertainty are addressed through an iterative, agile process involving the design, prototyping and testing of multiple variants (von Petersdorff-Campen, 2024a). Another tool used in product development is the so-called *Five Finger Formula,* where each “finger” represents key principle, namely (von Petersdorff-Campen, 2024b):

These key principles serve as a guideline for this project and are already evident in the formulation of the research questions, where each question represents an "isolated pain" as well as tests and metrics are addressed. The remaining "fingers" are going to be applied in the following description.

Supporting structure: "Foundation"

A vessel enclosing the substrate and its supporting structure form the basis of the apparatus. The given material requirements of cost-effectiveness and availability, as well as the simple functionalities, allow a 210-liter oil barrel cut lengthwise as a vessel and a wooden structure as an oil barrel rack. This implementation is likely to lead directly to the final components, dispensing with prototypes as intermediate steps.

The strength of the structure is to be assessed through mechanical stress tests.

Research Question 1: "Mixing"

A steel pipe is to function as the drive shaft and will be positioned concentrically along the longitudinal axis of the oil barrel. A mesh structure of reinforcing bars welded to the shaft shall form the agitator. The installation and removal of the combined agitator-shaft assembly are expected to be challenging due to geometric constraints. A potential solution is to divide the drive shaft at its centre and to reconnect it using a pipe coupling or a U-bolt clamp. After the drive shaft is mounted and supported by bearings, a hand crank will be attached to one end to manually actuate the agitator.

Given the low-cost and material availability constraints, the shaft bearing constitutes a challenge. In line with agile development, a quickly constructed, and low-cost wooden shaft bearing is to be designed and tested first. A standardised and expensive ball bearing is only to be implemented in the worst-case scenario.

The shaft bearing is checked by assessing how much force is required to turn the crank. According to the research question, the mixing quality is to be tested through visual inspection of photographs taken at different regions in the vessel and after varying numbers of crankshaft rotations.

Research Question 2: "Watering"

A water supply system shall ensure uniform moistening by evenly sprinkling the substrate. For this, a standard water tank will be mounted above the substrate container and will be connected to the water distribution system on the side wall of the oil barrel rack via a pipe system. The distribution system consists of a plastic sanitary pipe with several small holes drilled into it. Hydrostatic pressure created by the height difference between the water tank and the dispenser ensures a constant water flow through the boreholes.

The functionality of the water supply system will be tested as mentioned in the research question by applying a modified squeeze test described in (Oei, 2003, p.20) and by analysing the amount of recovered water from the substrate during compression.

Research Question 3: "Portioning"

A hole shall be cut at the bottom end of the oil barrel positioned horizontally on the rack. A slide valve shall control the connection to a filling cylinder below the opening. When the valve is open, substrate will fall into the cylinder, whose geometry defines the portion size. The filling cylinder is open on both ends and will run on a metal track. After the valve is closed, the cylinder will be moved along the track radially away from the oil barrel. An opening at the end of the track will allow the portioned substrate to fall into bags positioned below.

Tests, as described in the research question, will determine whether the implementation of a piston for further compression is required.

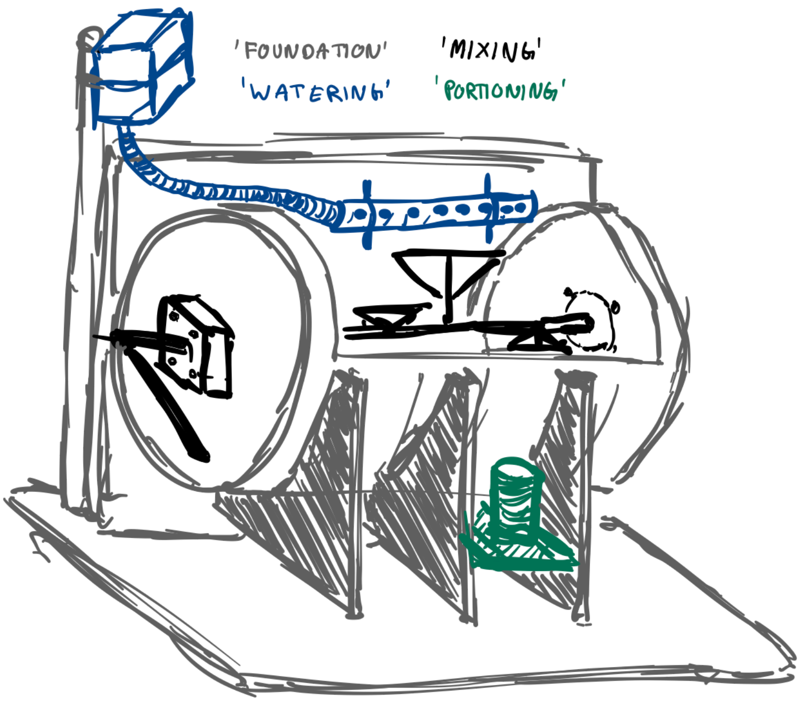

Figure 1: A first not scaled sketch as a rough overview of a possible product. Components can noticeably change due to agile development.

Planning

A **preparational phase** will be conducted to carry out background research, review relevant literature, and develop initial design concepts. This is followed by a brief **organisational phase** to procure the necessary tools and materials.

Subsequently, the **product development phases** are undertaken, comprising designing, manufacturing, testing, and documentation. These activities are performed concurrently to enable an agile development process and to minimize the risk of information loss that may occur if documentation is delayed. The **final phase** is used to combine already existing documentation of all production development phases into a written thesis.

Gantt chart

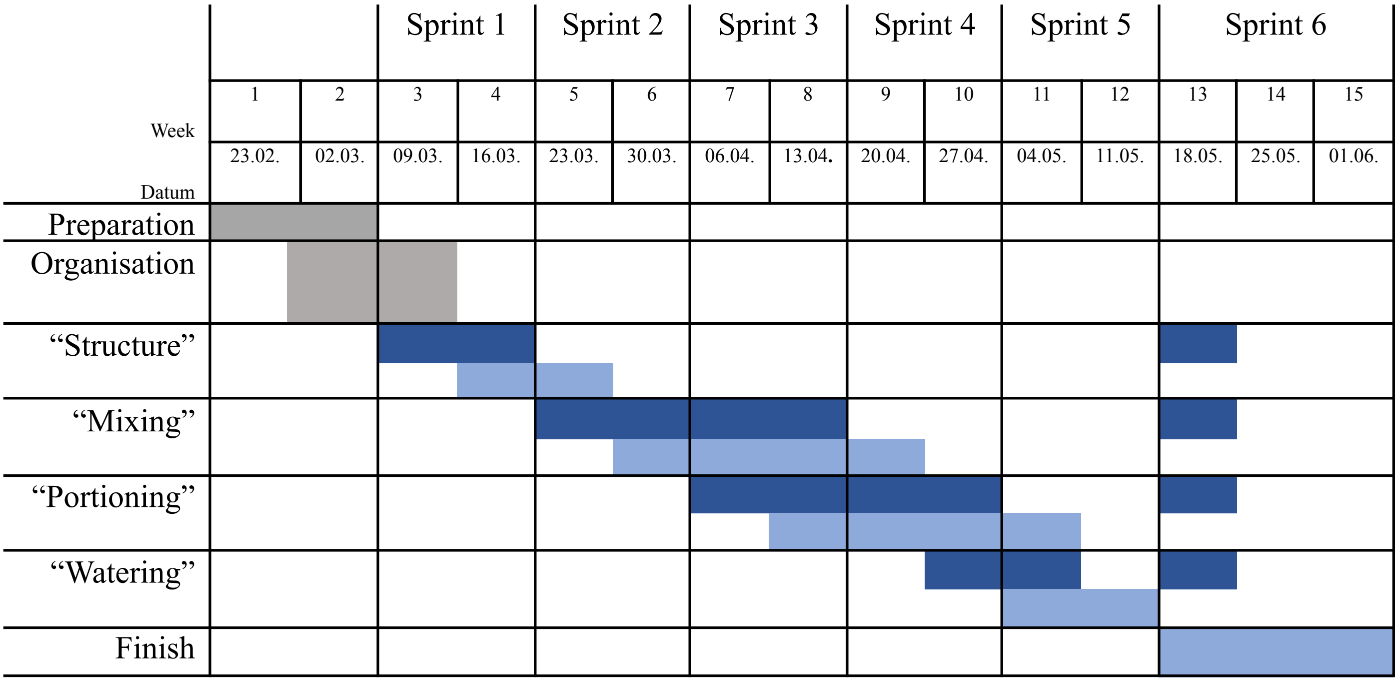

Figure 2. Gantt chart for describing the project's workflow. Preparatory and organisational phases are marked in grey. Product development phases are marked in blue, where manufacturing work appears darker, testing and documentation brighter.

As described in the method section is the product development carried out using the Scrum method and is therefore divided into six iterations called *Sprints* (Laloux, 2020). Each ending sprint should result in a complete system that can be used for substrate production. But every iteration is to refine the system, increasing its reliability and efficiency, and reducing the amount of manual work required for its use. At the conclusion of Sprint 1, a basic container is expected to be available, allowing mixing and portioning of the substrate through hand labour. By Sprint 5, a fully functional prototype should be completed, capable of reliably mixing, moistening, and portioning the substrate through mechanisms. Sprint 6 is reserved for further refinement and optimization, if necessary.

“*Mixing*” and “*Portioning*” are considered challenging due to their complex requirements, which is why more time is allocated for these tasks. Possible bottlenecks include the procurement of materials and tools. Progress will be delayed if, for example, no oil barrel or welding facility can be found during the organisational phase. The schedule has deliberately kept rough so that the focus can be flexibly adapted to potentially changing needs.

Data collection

dataset	data type	data format	data documentation
Structure test data"	*Observational data*	*Image and Video data (JPEG, MP4)* *Tabular data (CSV, XLSV)*	*Visually documented observation of stress tests by inducing force and moment in all three principal directions leading to photographs of the structure and a video showing manually performed stress tests. Axial and radial tilting moments shall be measured using a force gauge.*
Shaft bearing check	*Observational data*	*Tabular data (CSV, XLSX)*	*For each developed shaft bearing, the torque required to set the agitator in motion will be measured with a force gauge and supplemented with one of the following assessments of the effort required: "very low, low, moderate, high, very high"*
Substrate mixing test data	*Observational data*	*Image and Video data (JPEG, MP4)*	*Visually documented observation of mixing a substrate of various ingredients using the agitator.* *A measurement shall consist of a set of photographs of three samples taken after 0,10 and 20 crankshaft revolutions and repeated for mixing three substrates. (9 measurements, 27 photographs)* *Additionally, a video will illustrate the stirring process.*
Substrate portioning test data	*Observational data*	*Tabular data (CSV, XLSX)*	*Fill height and weight of the filling cylinder will be measured for 8 portions of substrate.* *Weight in g with 1g accuracy, height in mm with 5mm accuracy.*
Squeeze test data	*Laboratory data*	*Tabular data (CSV, XLSX)*	*Three moistened substrate samples of three test runs shall be equally pressurized by applying mass where the escaping water will be weighted (9 measurements total)* *Weight in g with accuracy to be adjusted to sample size.*
Reference parameters	*Secondary data*	*Tabular data (CSV, XLSX)*	*Literature values for reference in moisture, compactness and substrate composition.*

Deliverables

deliverable	data type	data format	data documentation
Device	*Hardware*	-	*Fully functional prototype of a substrate mixing, watering and portioning device.*
Assembly	*Modelled Data*	*3D CAD (STL)*	*3D CAD-model showing the device with all its components.*
Technical Drawings	*Modelled Data*	*Blueprints (PDF)*	*Technical drawing of each manufactured component.*
Assembly Instructions	*Modelled Data*	*(PDF)*	*Text and picture-based instruction for reconstructing the device.*

Data storage

The raw_data folder on Google Drive will be used to store all gained data. Observational data will be stored as pictures, videos or as tables. All data derived from raw data through testing will be stored in the testing folder. All stored data is to be annotated by supplementing test conditions like crankshaft revolutions, identification of sample, batch, prototype or sampling location.

Technical drawings of all components, assemblies and its instructions as well as CAD models will be stored in design and created using Autodesk Fusion 360 or equivalent.

Budget

No	expense	amount	unit price [CHF]	total price [CHF]	
1	Oil barrel 210l	2	80.00	0.00	gift Blaser Swisslube
2	Ball bearing	2	100.00	200.00	probably less expensive, maybe dispensable
3	Steel pipes	8m	2.50/m	20.00	discount Walker Haustechnik
4	Reinforcing steel	8m	5.00/m	0.00	gift Bärtschi Gebr. AG
5	Sanitary pipes	5m	3.00/m	15.00	discount Walker Haustechnik
6	Plastic container 20l	1	20.00	20.00	
7	Plywood plates	3	20.00	60.00	
8	Wooden beams	5m	4.00/m	20.00	
9	Small parts (screws…)	-	-	30.00	
11	Test Material (straw…)	-	-	50.00	
	Total			**415.00**	

This table should be regarded as a rough estimate, as no components have been purchased yet and material consumption may change at short notice due to agile development. A maximum budget of CHF 500 will be adhered to.

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